

Open-Vocabulary 4D Scene Understanding for Dynamic Driving Scenarios

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Abstract

We present a unified pipeline for open-vocabulary 4D scene understanding in dynamic driving environments. Our approach combines explicit 4D Gaussian-based dynamic reconstruction with point-level open-vocabulary instance understanding. Specifically, we integrate a Street Gaussians-style decomposition of static background and moving foreground objects with an OpenGaussian-inspired instance feature learning and language association framework. The resulting system aims to enable temporally coherent object discovery, segmentation, and text-based querying in complex urban scenes. We additionally explore the role of synthetic data augmentation for broader evaluation settings. On a Waymo Open Dataset perception scene, we observe an mIoU of 0.34 and a mean accuracy of 0.149 when comparing the predicted Gaussians’ centers with the ground truth point cloud.

1 Introduction

Understanding dynamic driving scenes requires both high-quality 4D reconstruction and semantic interpretation of objects over time. Recent 4D reconstruction methods can produce temporally consistent novel views but remain “semantically blind”, while open-vocabulary 3D Gaussian methods provide strong instance-level understanding but are primarily designed for static scenes. This gap limits downstream autonomous driving workflows such as long-tail event mining, rapid data annotation, and robust simulation.

This project aims to bridge this gap by combining explicit dynamic 4D scene reconstruction with open-vocabulary instance segmentation and language-based querying in a single framework. Our goal is to support object-centric temporal reasoning in real-world multi-view driving logs and to analyze performance under synthetically augmented variants of the data.

Contributions. Our main contributions are:

- A practical integration of explicit dynamic 4D Gaussian reconstruction with open-vocabulary instance feature learning for driving scenarios.
- A language association pipeline that enables zero-shot text-based object retrieval and tracking.
- An initial evaluation on Waymo Open Dataset scenes with quantitative reconstruction results and a plan for open-language instance metrics.

4D Dynamic Reconstruction. Feed-forward and optimization-based approaches have advanced large-scale spatio-temporal reconstruction in outdoor scenes. Methods such as STORM demonstrate temporally consistent 4D reconstruction using Gaussian representations and self-supervised aggregation mechanisms [15].

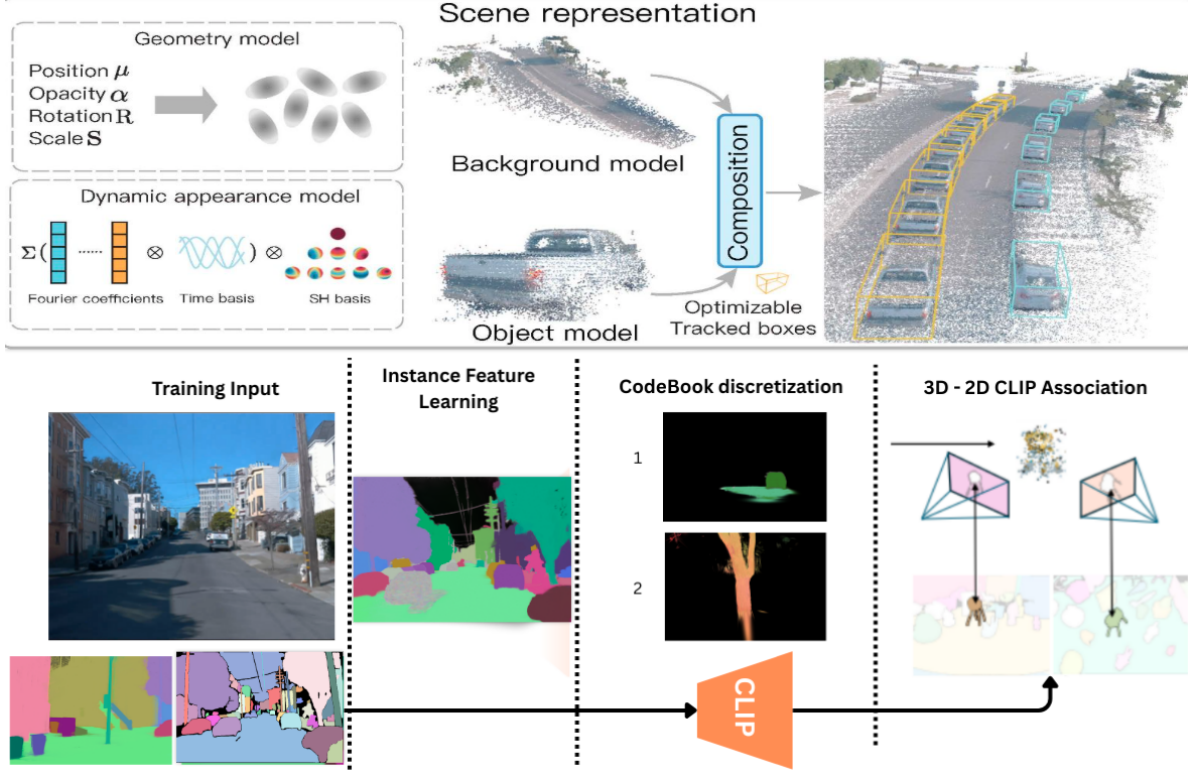


Figure 1: Overview of the proposed methodology pipeline. We integrate explicit dynamic scene decomposition (Stage 1) with open-vocabulary semantic understanding. The pipeline learns continuous 3D instance features supervised by 2D SAM masks using smoothing ($\mathcal{L}_s$) and contrastive ($\mathcal{L}_c$) losses (Stage 2). These features are discretized via a two-level codebook strategy (Stage 3) and finally associated with CLIP language embeddings to enable zero-shot text-based querying and segmentation of dynamic objects (Stage 4).

Explicit Static-Dynamic Decomposition. Hybrid approaches that separate time-independent background from time-dependent moving objects simplify dynamic rendering and enable object-level manipulation. Our pipeline adopts this explicit decomposition to isolate moving vehicles from static context, inspired by recent street-scale Gaussian approaches [14].

Open-Vocabulary 3D Gaussian Understanding. OpenSplat3D and OpenGaussian introduce learnable instance embeddings supervised by 2D masks (e.g., from SAM), enabling zero-shot semantic identification of 3D instances [5, 13, 2]. However, these methods focus primarily on static scenes. We extend similar feature learning principles into a dynamic driving setting.

Text-Guided Novel Video Synthesis. Recent work has explored temporally and multi-view consistent generation of driving scenes using intermediate representations such as HD maps. We incorporate this direction as a dataset augmentation pathway to expand the diversity of evaluated scenarios [7, 11, 4].

2 Methodology

Our approach integrates explicit dynamic scene representation with open-vocabulary semantic understanding. We adopt a unified pipeline that first reconstructs the dynamic urban environment using the *Street*

Gaussians framework and subsequently augments this representation with distinct, language-grounded instance features via *OpenGaussian*.

2.1 Dynamic Scene Representation

To address the challenges of rendering large-scale dynamic urban streets, we model the scene as a composition of explicit point-based sets. The scene is decomposed into a static background model $\mathcal{M}_b$ and a set of moving foreground object models $\{\mathcal{M}_o^i\}_{i=1}^N$.

2.1.1 3D Gaussian Parameterization

Each point in these models is parameterized by a 3D Gaussian, defined by a center position μ , covariance matrix Σ , opacity α , and color parameters c . The influence of a Gaussian at a spatial location x is defined as:

$$G(x|\mu, \Sigma) = \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right) \quad (1)$$

To ensure the covariance matrix Σ remains positive semi-definite during optimization, it is decomposed into a rotation matrix R and a scaling matrix S such that $\Sigma = RSS^T R^T$.

2.1.2 Object Tracking and Transformation

Unlike the background, which is defined in the world coordinate system, foreground vehicles are defined in a local object coordinate system. To render the scene at time t , local object Gaussians are transformed into the world space using optimizable tracked poses. Given a tracked rotation R_t and translation T_t , the world position μ_w and covariance Σ_w are computed as:

$$\mu_w = R_t \mu_o + T_t, \quad \Sigma_w = R_w S_o S_o^T R_w^T \quad (2)$$

where $R_w = R_t R_o$, μ_o is the position in the object frame, and S_o is the object scaling matrix. We treat the tracked poses (R_t, T_t) as learnable parameters, optimizing them jointly with the Gaussian attributes to compensate for tracking noise.

2.1.3 4D Spherical Harmonics for Dynamic Appearance

Moving vehicles exhibit time-varying appearance changes, such as shifting specular highlights and shadows, which static Spherical Harmonics (SH) cannot capture. We employ a 4D SH model where the coefficients are time-dependent. We model each SH coefficient $z_{m,l}$ using a Fourier series expansion:

$$z_{m,l}(t) = \sum_{k=0}^{K-1} \mathbf{f}_k \cos\left(\frac{k\pi}{N_t} t\right) \quad (3)$$

Here, $\mathbf{f}_k$ represents the learnable Fourier coefficients and N_t is the total time duration. This allows the model to interpolate complex lighting changes smoothly over time.

2.2 Continuous Instance Feature Learning

To extend this geometric representation with semantic understanding, we augment each 3D Gaussian with a learnable low-dimensional feature vector $f \in \mathbb{R}^6$. We utilize 2D masks generated by the Segment Anything Model (SAM) for supervision. To achieve 3D consistency without expensive cross-frame tracking, we optimize the features using two specific loss functions.

Let $M \in \mathbb{R}^{6 \times H \times W}$ be the rendered feature map for a given view. For a specific binary mask B_i , let $\bar{M}_i$ denote the mean feature vector of all pixels falling within that mask.

2.2.1 Intra-mask Smoothing Loss

To ensure that all Gaussians belonging to the same object instance converge to a unified feature representation, we minimize the feature variance within each predicted 2D mask (using L2 distance):

$$\mathcal{L}_s = \sum_{i=1}^m \sum_{(h,w) \in B_i} \|M_{h,w} - \bar{M}_i\|_2^2 \quad (4)$$

This loss forces the rendered features $M_{h,w}$ at pixel coordinates (h, w) to align closely with the instance prototype $\bar{M}_i$.

2.2.2 Inter-mask Contrastive Loss

To ensure features are distinct across different objects, we maximize the distance between the mean features of different masks i and j :

$$\mathcal{L}_c = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j \neq i} \frac{1}{\|\bar{M}_i - \bar{M}_j\|_2^2} \quad (5)$$

This contrastive learning objective pushes separate instances to form orthogonal clusters in the feature space, ensuring distinctiveness even for objects of the same semantic class.

2.3 Discrete Instance Abstraction

Continuous features can suffer from noise and ambiguity. To resolve this, we employ a **Two-Level Code-book** strategy to discretize the learned features to ensure that Gaussians from the same instance possess identical (not just similar) features.

1. **Coarse Level:** We perform clustering based on a concatenated vector of the semantic feature f and the spatial position x_{pos} (i.e., $[f; x_{pos}]$). This is crucial for separating spatially distant objects that may share similar textures or colors.
2. **Fine Level:** Within each coarse cluster, we perform a second refinement based solely on the semantic feature f to sharpen object boundaries.

2.4 Language-Grounded Association and Temporal Tracking

Finally, we enable open-vocabulary queries by associating the 3D instances with text descriptions via CLIP. We propose a lightweight matching mechanism that avoids the need for depth rendering or occlusion testing.

For a rendered feature map of a specific 3D instance M_{inst} and a candidate 2D SAM mask B_{sam} , we compute a unified matching score S :

$$S = \text{IoU}(\pi(M_{inst}), B_{sam}) \cdot (1 - \|\bar{M}_{inst} - P_{sam}\|_1) \quad (6)$$

where IoU measures the spatial overlap, and the second term penalizes the L_1 feature distance between the instance and the feature-filled mask P_{sam} . The 3D instance is assigned the CLIP embedding of the 2D mask that maximizes this score.

Text-Based Selection and Tracking: This language association allows users to select objects using natural language queries (e.g., “red sports car”). Our framework uniquely enables *temporal tracking* of these selections by leveraging the decomposed nature of Street Gaussians. Once an object is selected via OpenGaussian’s feature matching, the selection is bound to the specific foreground object model $\mathcal{M}_o^i$. Since $\mathcal{M}_o^i$ is explicitly driven by the continuous tracked poses R_t and T_t (from Eq. 2), the semantic selection is automatically and robustly propagated across the entire temporal sequence without requiring frame-by-frame re-identification.

2.5 Dataset Augmentation with Text-Guided Novel Views

We explore synthetic augmentation of multi-view driving sequences in order to test the working of the 4D pipeline on synthetic data. The pipeline first produces intermediate HD map-like representations, which can guide multi-view prompt-controlled synthesis in a temporally consistent manner. This provides a route to expand semantic diversity without collecting new real-world data. Recent work from NVIDIA [7] allows us to make consistent modifications to scenes using text-based prompts.

3 Experiments

3.1 Dataset

We conduct experiments on the Perception subset of the Waymo Open Dataset [9], which contains diverse multi-camera urban driving sequences captured from moving vehicles. This setting is well-suited for evaluating both dynamic reconstruction fidelity and open-vocabulary instance understanding.

3.2 Evaluation Protocol

To evaluate open-vocabulary understanding, we assign each 3D Gaussian cluster a text label by comparing cosine similarity between label CLIP embeddings and Gaussian text features. We compute IoU and classification accuracy by comparing Gaussian-associated points against semantically segmented point clouds provided by Waymo.

4 Results

4.1 Quantitative Observations

The OpenWaymo Perception dataset provides semantic labels of 23 classes for its 3D point clouds at a rate of 2Hz, or approximately 20% of all frames. For a given frame, we calculate mIoU and accuracy with point clouds by matching each predicted Gaussian’s center with the nearest point in the ground truth. We calculate mIoU and mean accuracy using the prediction-ground truth pairs of points. We measure a mIoU of 0.34 and a mean accuracy of 0.149 on our test frame.

4.2 Qualitative Observations

Qualitative renderings show:

- Clear separation between static background and moving foreground.
- Temporally stable appearance of tracked vehicles under 4D SH modeling.



Figure 2: Qualitative visualization of the 4D vehicle. The result illustrates clear separation of dynamic cars from the static background and supports object-centric analysis.

- Promising instance feature maps that align with object boundaries, forming the basis for language-grounded object selection.

4.3 Dataset Augmentation Outputs

The following figures show the intermediate and final outputs obtained when augmenting the dataset with synthetically generated scenes.

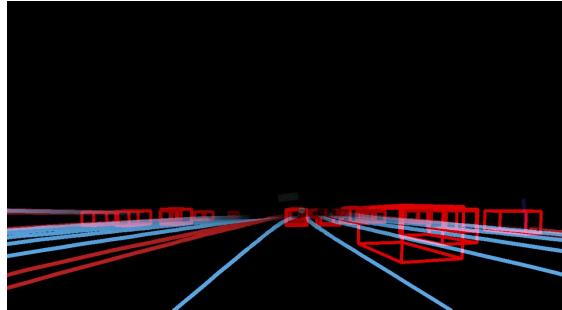


Figure 3: A still from the intermediate HD map output video obtained by using the Cosmos Drive Dreams model.



Figure 4: Stills from the final outputs of the Cosmos Dream Drive model with realistic scene modifications, while maintaining semantic structure.

5 Discussion, Limitations, and Challenges

Our results suggest that combining explicit 4D reconstruction with open-vocabulary instance features is a viable path toward object-centric temporal understanding in dynamic driving scenes. The reconstruction metrics and qualitative renderings indicate that static–dynamic decomposition and pose-tracked foreground modeling can produce stable, high-fidelity results suitable for downstream instance-aware analysis.

However, several limitations and practical challenges emerged during implementation and evaluation:

- **Limited compositional mask supervision.** A key bottleneck was the lack of an off-the-shelf model that could reliably produce instance masks conditioned on *compositional* natural language descriptions such as “*red car turning right*” or “*pedestrians walking left*”. In practice, the segmentation tools we tested primarily supported category-level prompts (e.g., “*car*”, “*pedestrian*”), which reduced supervision to coarse object-level detection rather than fine-grained attribute- or motion-aware instance parsing. This constrained our ability to learn features that distinguish objects by appearance and action, and limited text-to-instance grounding to broad class retrieval rather than richer phrase-level queries.
- **Ambiguity in dynamic scenes and mask noise.** Urban driving scenes often contain multiple similar instances in close range, with partial occlusions and rapid viewpoint changes. Under category-only supervision, nearby instances can be merged or inconsistently segmented across frames, increasing noise during instance feature optimization.
- **Scalability of instance discretization.** Large scenes with dense traffic may require more robust hierarchical clustering or memory-efficient discretization strategies beyond our current coarse-to-fine codebook setup.
- **Compute and pipeline complexity.** Integrating 4D reconstruction, pose-tracked foreground modeling, mask-guided feature learning, and discretization introduced a multi-stage dependency chain. This increased iteration time and made it difficult to exhaustively evaluate all combinations of hyperparameters and codebook settings within project constraints.
- **Synthetic augmentation cost.** Multi-view, temporally consistent text-guided generation ended up being a major challenge due to the large compute resources that were required for it. Thus, in order to work around this issue, modifications had to be made to the inputs (eg. scene length), to make sure inference was possible on smaller GPUs.

6 Future Work

We see three high-impact directions.

Object-Centric 4D World State. We plan to cluster 4D Gaussians into 3D instances while maintaining temporally consistent IDs. This would enable per-object trajectories, an object–time table (or 4D scene graph), and higher-level motion descriptors such as velocity, heading, lane changes, and collisions.

Language-Programmable Temporal Queries. A structured object–time representation can allow grounding of natural-language queries into executable constraints over object attributes and motion descriptors, supporting retrieval of objects, trajectories, and interaction events.

Expanded Augmented Evaluation. We aim to generate additional prompt-guided, multi-view consistent scenes to quantify robustness across a wider distribution of lighting, weather, and long-tail semantic configurations.

7 Conclusion

We presented an integrated pipeline for open-vocabulary 4D scene understanding in dynamic driving scenarios. By combining explicit static–dynamic 4D Gaussian reconstruction with OpenGaussian-style instance feature learning, coarse-to-fine discretization, and CLIP-based language association, our approach moves toward temporally grounded, text-interactive 4D world interpretation. Initial results on Waymo scenes show promising reconstruction quality, and our planned open-language evaluations and augmented-data experiments provide a clear path for extending the framework.

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